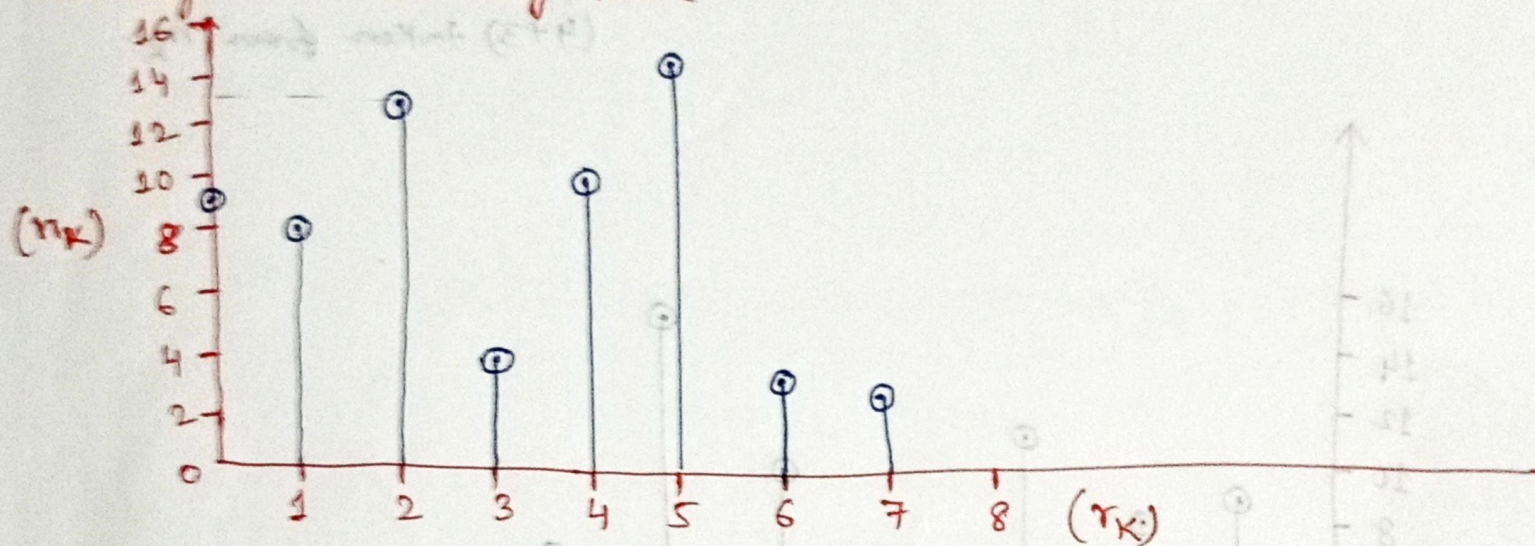


① Perform Histogram Equalization for an 8x8 Image.

Gray Levels	0	1	2	3	4	5	6	7
Number of pixels	9	8	11	4	10	15	4	3

Gray levels provide intensity values and number of pixels having that intensity value.



Gray Levels (r_k)	No. of pixels (n_k)	$P(r_k) = \frac{n_k}{n}$ (PDF)	S_k (CDF)	$S_k * 7$ (8 levels - 1)	Histogram Equalization level (Round off $S_k * 7$)
0	9	$9/64 = 0.141$	0.141	$0.141 \times 7 = 0.987$	1
1	8	$8/64 = 0.125$	$0.141 + 0.125 = 0.266$	1.862	2
2	11	$11/64 = 0.172$	$0.266 + 0.172 = 0.438$	3.066	3
3	4	$4/64 = 0.0625$	$0.438 + 0.0625 = 0.5005$	3.5035	4
4	10	$10/64 = 0.156$	$0.5005 + 0.156 = 0.6565$	4.5955	5
5	15	$15/64 = 0.234$	$0.6565 + 0.234 = 0.8905$	6.2335	6
6	4	$4/64 = 0.0625$	$0.8905 + 0.0625 = 0.953$	6.671	7
7	3	$3/64 = 0.047$	$0.953 + 0.047 = 1$	7	7

Updated Histogram Table

Gray Levels	1	2	3	4	5	6	7
No of pixels	9	8	11	4	10	15	7

7 appears twice so
(4+3) taken from n_k .

